Radiotherapy and Oncology 166 (2022) 15–25

Contents lists available at ScienceDirect

Radiotherapy and Oncology

journal homepage: www.thegreenjournal.com

Systematic Review

Local tumor control and treatment related toxicity after plaque brachytherapy for uveal melanoma: A systematic review and a data pooled analysis

Francesca Buonanno [a], Manuel Conson [b], Cintia de Almeida Ribeiro [c], Caterina Oliviero [d], Francesca Itta [a], Raffaele Liuzzi [e], Roberto Pacelli [b], Laura Cella [e,*,1], Stefania Clemente [d,1]

[a] University Federico II, Post Graduate School in Medical Physics, Department of Advanced Biomedical Sciences; [b] University Federico II, Department of Advanced Biomedical Sciences, Napoli, Italy; [c] Brazilian Institute of Cancer Control, San Paolo, Brazil; [d] University Hospital Federico II, Unit of Medical Physics and Radioprotection; and [e] National Research Council (CNR), Institute of Biostructures and Bioimaging, Napoli, Italy

ARTICLE INFO

Article history:
Received 14 July 2021
Received in revised form 29 October 2021
Accepted 2 November 2021
Available online 11 November 2021

Keywords:
Uveal melanoma
Eye plaque brachytherapy
Local tumor control
Ocular morbidity
TCP
NTCP

ABSTRACT

Uveal melanoma (UM) represents the most common primary intraocular tumor, and nowadays eye plaque brachytherapy (EPB) is the most frequently used visual acuity preservation treatment option for small to medium sized UMs. The excellent local tumor control (LTC) rate achieved by EPB may be associated with severe complications and adverse events. Several dosimetric and clinical risk factors for the development of EPB-related ocular morbidity can be identified. However, morbidity predictive models specifically developed for EPB are still scarce.

PRISMA methodology was used for the present systematic review of articles indexed in PubMed in the last sixteen years on EPB treatment of UM which aims at determining the major factors affecting local tumor control and ocular morbidities. To our knowledge, for the first time in EPB field, local tumor control probability (TCP) and normal tissue complication probability (NTCP) modelling on pooled clinical outcomes were performed.

The analyzed literature (103 studies including 21,263 UM patients) pointed out that Ru-106 EPB provided high local control outcomes while minimizing radiation induced complications. The use of treatment planning systems (TPS) was the most influencing factor for EPB outcomes such as metastasis occurrence, enucleation, and disease specific survival, irrespective of radioactive implant type. TCP and NTCP parameters were successfully extracted for 5-year LTC, cataract and optic neuropathy. In future studies, more consistent recordings of ocular morbidities along with accurate estimation of doses through routine use of TPS are needed to expand and improve the robustness of toxicity risk prediction in EPB.

Uveal melanoma (UM) represents the second most common form of melanoma and the most common primary intraocular tumor with an occurrence, in both the U.S. and Europe, of about 5–7.5 cases per million people per year [1–4].

UMs typically appear as dome shaped mounds, which involve primarily the choroid (90%) and to a lesser extent the ciliary body and the iris (6% and 4%, respectively) [5]. According to UM size and location, diverse treatment choices exist such as enucleation, stereotactic radiotherapy, charged particle irradiation and eye plaque brachytherapy (EPB) [6]. Up to '80s, the treatment of choice for UMs was enucleation. However, for small and medium sized UMs, a large trial from the Collaborative Ocular Melanoma Study (COMS) established no differences in survival between patients undergoing enucleation compared to EPB treatment [7]. With EPB, an eye applicator (the plaque) containing a radioactive source (gamma or beta emitter) is surgically affixed beneath the tumor base in direct contact with the ocular bulb for the time necessary for the prescribed dose to reach the tumor apex. In clinical practice, different radioisotopes are used including low-energy gamma radioisotope (I-125 and Pd-103) seeds along with the beta-emitting Ru-106 and Sr-90 plaques. Traditionally, the most common treatment in North America is I-125 EPB while Ru-106 is the isotope of choice in Europe and Asia [8,9]. Ir-192, Cs-131, Au-198 and miscellaneous loaded plaques are also used worldwide [10]. The excellent tumor control rate of over 90% achieved by EPB [11,12] can however be associated with severe complications and adverse events. A wide

* Corresponding author at: National Research Council (CNR), Institute of Biostructures and Bioimaging, Napoli, 80141, Italy.

E-mail address: laura.cella@cnr.it (L. Cella).

[1] These authors share the senior authorship.

https://doi.org/10.1016/j.radonc.2021.11.007